

Computer Networks

Lecture 10: Data Link Layer

Content



- **Framing**

Example

Find the CRC bits if the data word is **110101** and the generator polynomial is **1011**

- Original Data: 110101
- After appending zeros: **110101000**

110101000 ÷ 1011

1011

1100

1011

1111

1011

1000

1011

01100

1011

0111

- Remainder: 111
- Codeword: 110101111

Question 1

A data word is **1101011011**. The generator polynomial is $G(x) = x^4 + x + 1$

- Find the CRC bits.
- Form the transmitted codeword.
- Verify whether the receiver detects an error if the received word is **110101101100100**.

Question 2

Given message **11100011** and generator polynomial $= x^3 + x^2 + 1$

- Compute CRC bits.
- Form the codeword.
- Show that single-bit error is detected.

Question 3

Why After Appending CRC Remainder We Get ZERO Remainder?

Let dataword to be transmitted is $D(x)$

In order to convert it into codeword we append some Zeros. Let it be $D(x).x^r$

Let $G(x)$ be the generator function thus

$$D(x).x^r / G(x) \Rightarrow Q(X), R(x)$$

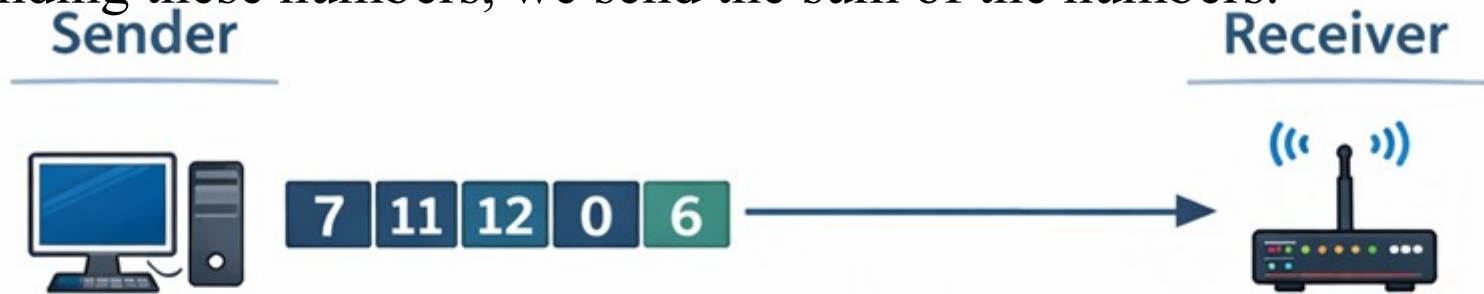
So mathematically, $D(x).x^r \Rightarrow Q(x).G(x) + R(x)$

Transmitted Data: $D(x).x^r + R(x) \Rightarrow Q(x).G(x) + R(x) + R(x)$

What is $R(x) + R(x)$?

Checksum

Suppose our data is a list of five 4-bit numbers that we want to send to a destination. In addition to sending these numbers, we send the sum of the numbers.



No Error

$$7 \ 11 \ 12 \ 0 \ 6 = 36$$

$$7 + 11 + 12 + 0 + 6 = 36$$



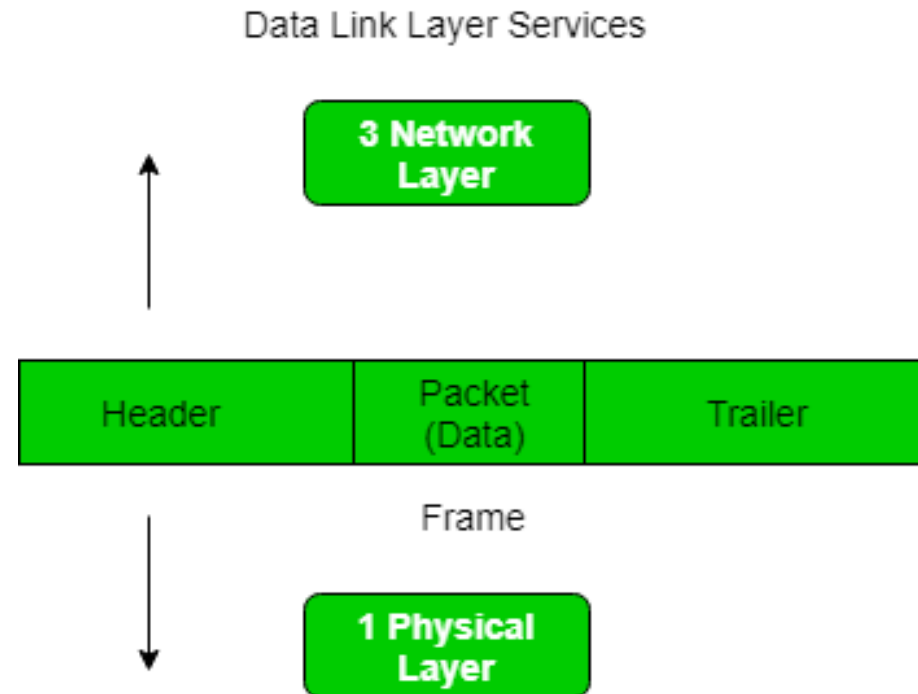
Accept, discard sum

Framing

Framing in the data link layer is the process of encapsulating network layer packets into manageable units called frames, adding header (addresses) and trailer (error detection) bits for reliable transmission.

Although the whole message could be packed in one frame, that is not normally done.

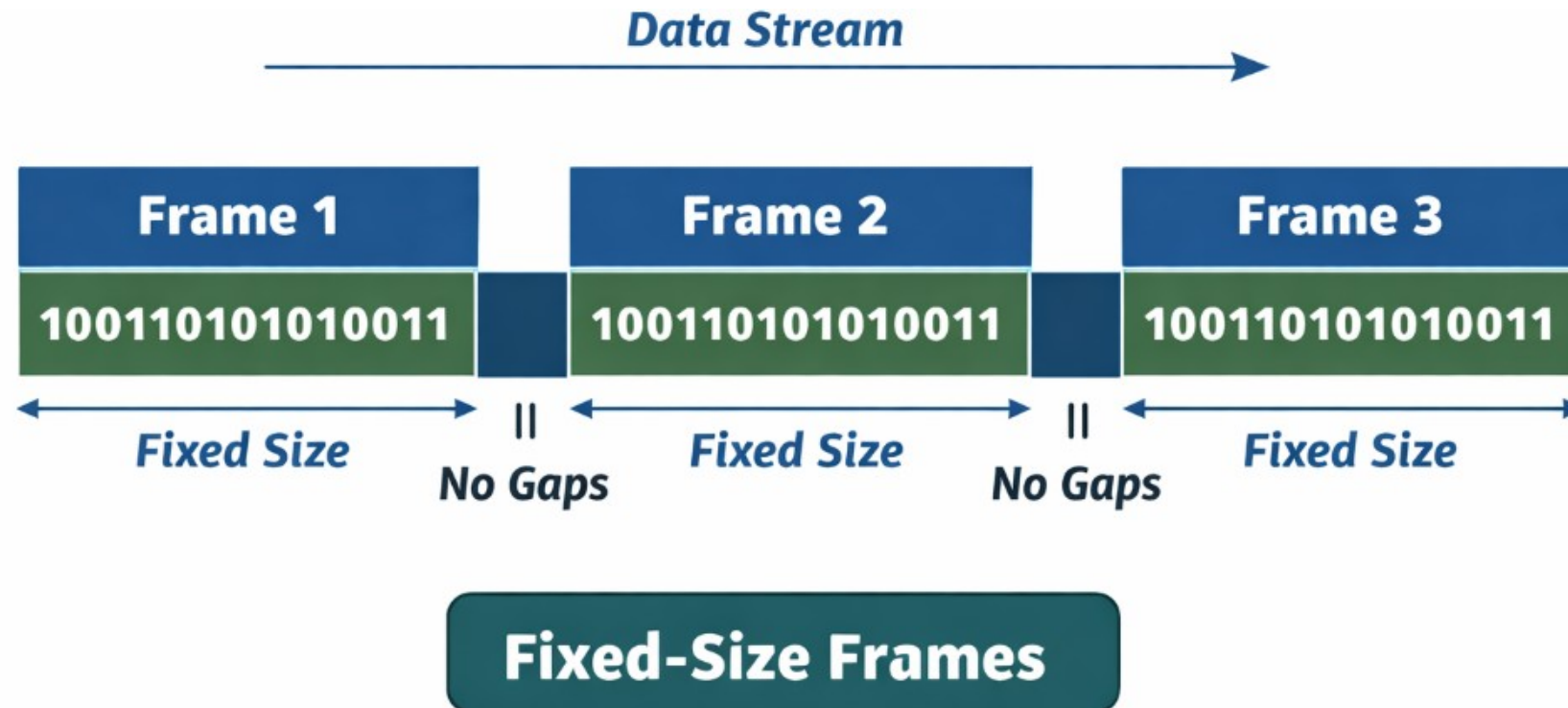
One reason is that a frame can be very large, making flow and error control very inefficient.



Fixed Size Framing

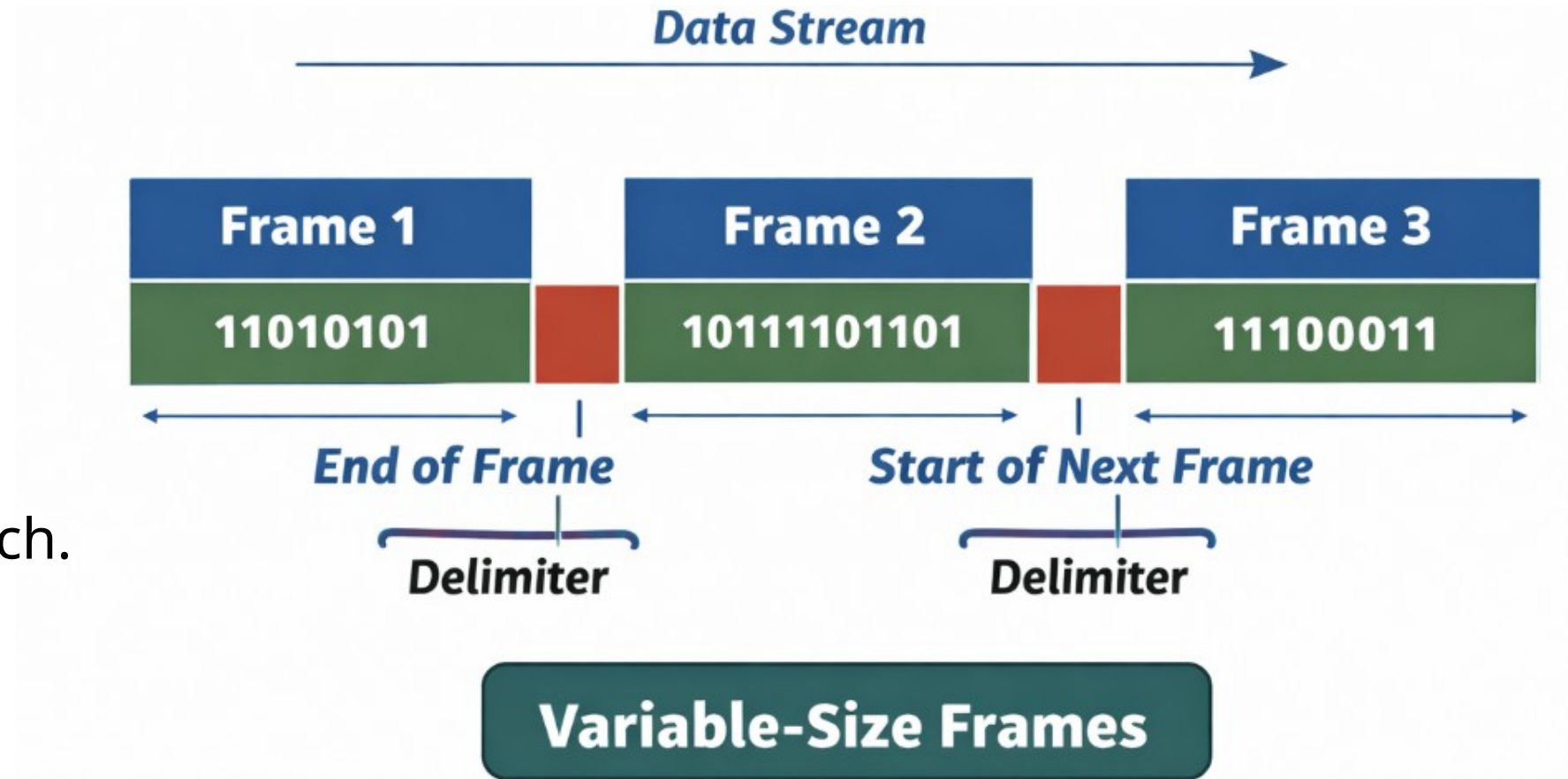
Frames can be of fixed or variable size.

In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.



Variable size Framing

- A character-oriented approach
- A bit-oriented approach.

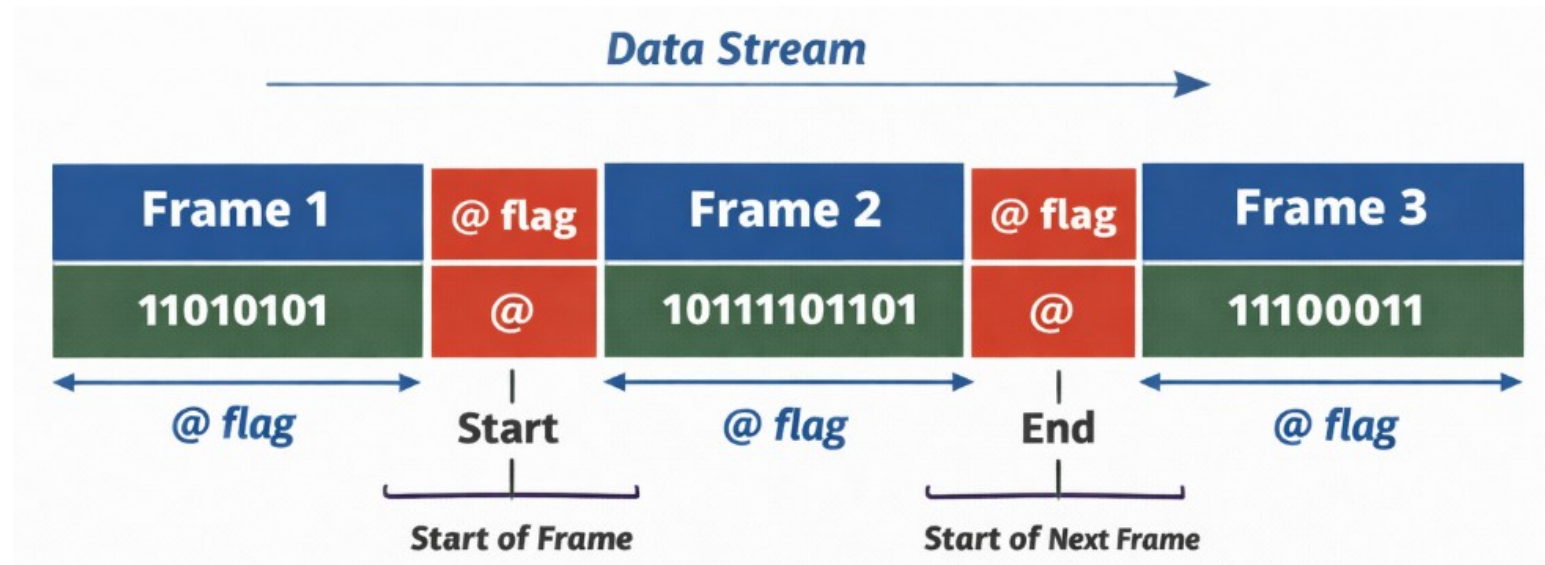


Variable size Framing

Character-Oriented Protocols

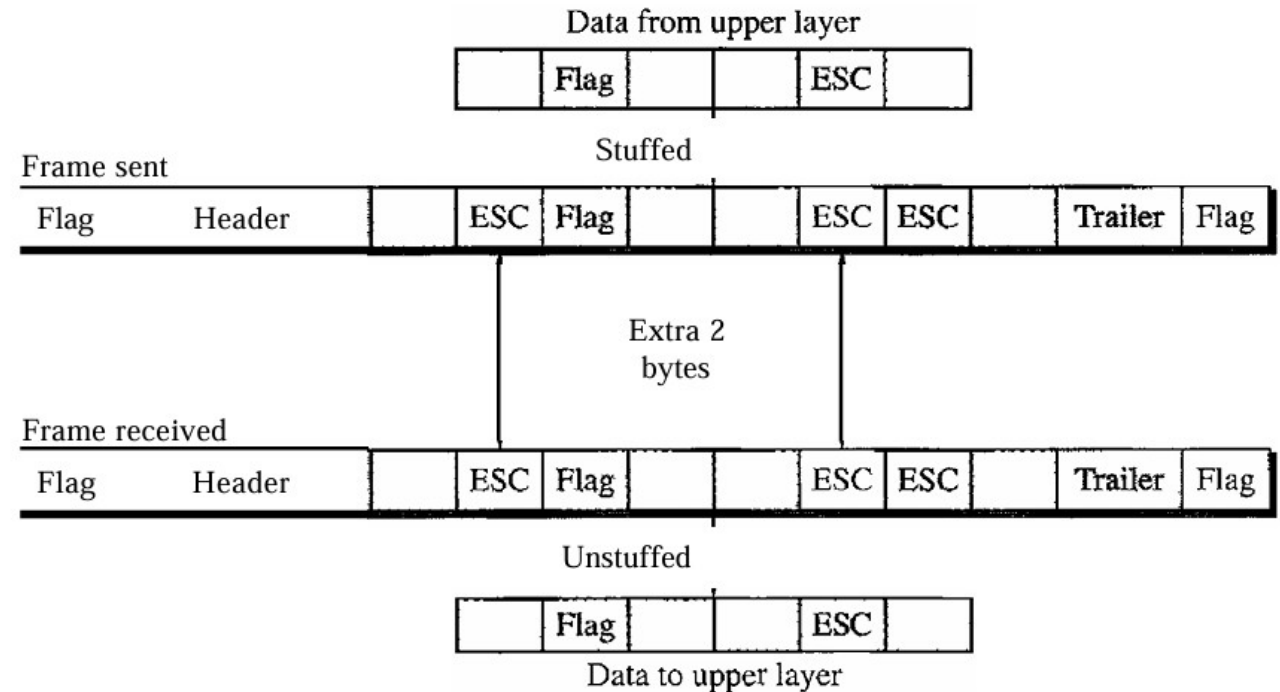
In a character-oriented protocol, data to be carried are **8-bit** characters from a coding system such as *ASCII*

To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.



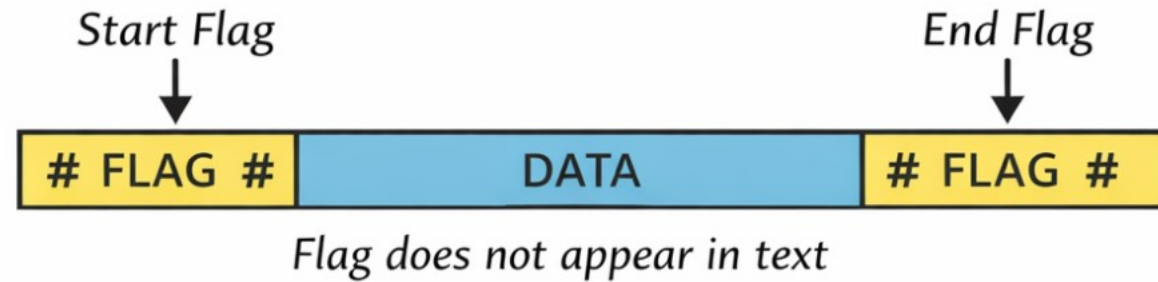
Character Oriented Framing

- Character-oriented framing was popular when only text data was transmitted.
- A special flag character was used to mark the start and end of a frame.
- The flag was chosen such that it did not appear in normal text communication.



Character Oriented Framing

- Any bit pattern used as a flag may appear in the actual data.
- If the receiver encounters the flag pattern inside data, it incorrectly assumes the frame has ended.



Example:



- # = Frame Delimiter
- HELLO WORLD = Actual Data

Incorrect Case:



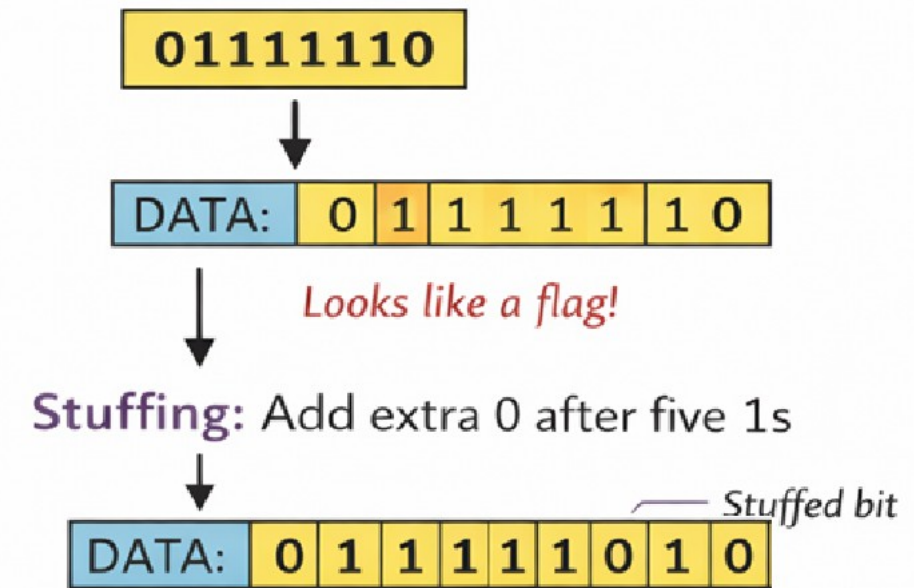
Bit Oriented Framing

- Data is transmitted as a sequence of bits (text, audio, video, graphics).
- Frames need a delimiter to mark start and end.
- A special 8-bit flag pattern: 01111110 is used as delimiter.
- The flag separates one frame from the next.
- If the flag pattern appears in data, bit stuffing is used to avoid confusion.

| 01111110 | Header | DATA | Trailer | 01111110 |

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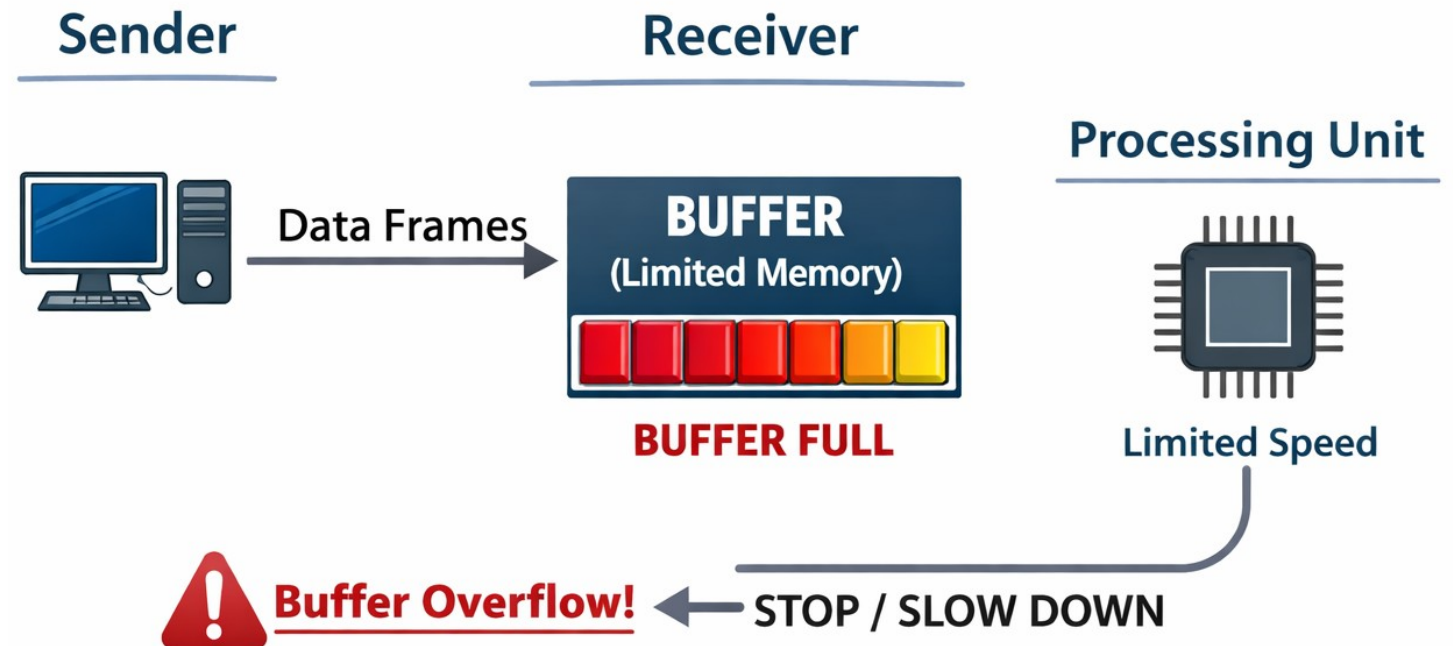
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Flow and Error Control

Flow control coordinates how much data the sender can transmit before receiving acknowledgment. Prevents the sender from overwhelming the receiver.

- Limited Processing Speed
- Limited Buffer Memory

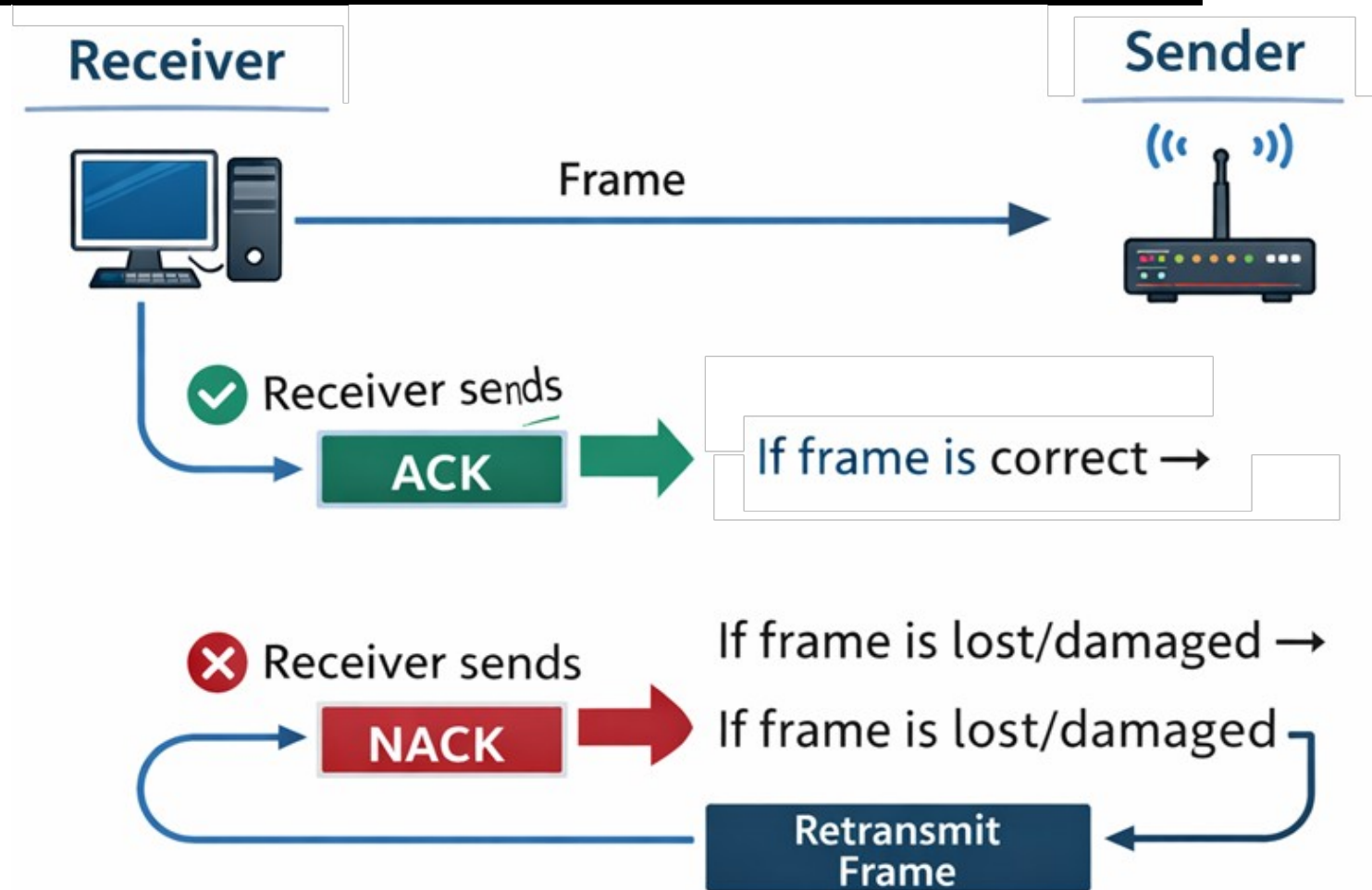
- Buffer stores incoming frames temporarily
- If buffer fills → Receiver tells sender to **stop or slow down**



Flow and Error Control

Error control deals with

- Detecting errors
- Correcting errors
- Retransmitting lost/damaged frames



Data Link Control Protocol

Value of parity bit is such that character has even (even parity) or odd (odd parity) number of ones

